

Cover letter

We are pleased to provide the selected components and equipment based on the following application information.

Product: wood chips 3/8" minus cross section

Incoming Moisture Content: 50%

Est Bulk Density: 10 lb/ft³

Desired Output MC: 10%

Lb/hr in: 8,650

Lb/hr out: 4805

Lb of water removed 3,844.43

Gal of H₂O removed 460.67

Est BTU of entire system 6,727,777 BTU

Fuel source: Biomass Burner

Executive summary



Quote

Quote ref.	20260730-131502189
Issue date	Jul 30, 2026
Expires	Oct 28, 2026
Currency	USD

Seller

Midwest Custom Engineering, Inc.

6526 S Kanner Hwy #215

Stuart, FL 34997

United States

Contact

Jason Shipley (jason@usemce.com)

+16202009109

Buyer

Nix Forest Industries

14732 State Hwy 87

Timpson, TX 75975

Contacts

Jim Nix (jimnix911@yahoo.com)

Effective date	Total discount	Total contract value
On agreement	\$0.00	\$786,761.00

Line items

Name	Quantity	Unit price	Unit discount	Net price
XD-72 Mk.2 Rotary Thermal Conditioning System	1	\$489,000.00	—	\$489,000.00

Rotary Thermal Conditioning System - BUDGETARY PROPOSAL

Buyer to provide clean, sized product with the following specifications:

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XD Series - Heavy Duty Direct Fired Rotary Thermal Conditioning System

- Feed Airlock system with 2 HP motor
- Co-current single Pass Rotary Dehydration Drum (6' dia x 35' long). Drum to be constructed from A36 steel.
- Heavy Duty Dryer Drop-Out Box for dry product
- Airlock complete with 1.5 HP motor, drive and control devices, Mild Steel construction.
- Drum Drive mounted on a structural base complete with 20 HP motor and drive.
- Full tooth, mounted Chain Sprocket to match chain drive.
- Forged steel Drum Tracks with HD attachment points
- Front Trunnion Base with Hardened Trunnion Wheels mounted on a structural base.
- Rear Trunnion Base with Hardened Trunnion Wheels and Thrust Wheel Assembly mounted on a structural base.
- Front and Rear Water Deluge Nozzle for fire suppression.
- Inlet and Discharge Seals.

Exhaust Fan

- XF-36 Fan
- Arrangement #9F heavy-gauge steel base with continuous welds.
- Heavy-duty pillow block bearings.
- Turned, ground, and straightened shaft.
- Dynamically and statically balanced rotor with tapered hub and bushing.
- Narrow width high static, heavy-duty rotor.
- Rotor to be fabricated of AR 235 Plate Steel.
- Heavy duty housing with continuous welds.
- Housing to be fabricated of carbon steel.
- Motor slide base with integral drive tensioning mechanism.
- 60 HP, 1800 RPM, class TEFC, 230-460V/3/60 NEMA Premium efficient motor.
- NOTE: TEFC motor with no environmental marking/T Code.
- V-belt drive with 1.5 to 2.0 service factor.

Name	Quantity	Unit price	Unit discount	Net price
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- Totally enclosed sanitary belt and shaft guard.
- Factory test run prior to shipment.
- Inlet to 21" OD.
- Flanged rectangular outlet.
- Vibration isolation sleeve with clamps for fan inlet.
- Bolt-on inspection door.
- Housing to have a drain port in the bottom.
- Product contact zones unpainted, all other surfaces painted Basalt Gray; guards painted safety orange. Please specify otherwise.

Transfer System

- Not quoted at this time

Ducting and Transition

- Ducting, Transitions and Elbows up to 60 lineal feet, A36 construction.

Control System and Fire Prevention

- Basic Burner Management System Control Panel
- Please add an additional \$29,000 if PLC system with up to 1 man weeks programming and HMI Graphics .

Additional programming at buyers expense

- Electrical and Control Instrumentation for MCS's proprietary processing equipment.
- Motor Control Cabinet for Seller's equipment , 3 Phase , 60 Hertz , 480 Volt.
- Manual quench nozzles provided - automated fire suppression system quoted upon request

System Commissioning

- 1 man-weeks required

Equipment Installation Instructions for Buyer's installation contractors.

- Start-up and Operator Training

Billing summary: Starts on effective date

6.5MM BTU/hr Solid Fuel Biomass Burner	1	\$264,285.00	—	\$264,285.00
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6.5MM BTU/hr Solid Fuel Biomass Burner

1 ea. New 6.5MM BTU/hr Solid Fuel Biomass Burner - for Drum Dryer

Solid fuel burner is lined with an excellent combination of refractories to assure best heat retention and long burner life. The stack and hot gas transfer duct are lined with a high-temperature light-weight insulating refractory. All steel surfaces are painted with a high-temperature anti-oxidant paint. Excess air fans, solid fuel feed hopper, and fuel feed fan with 10 feet of fuel feed pipe are included. Also included is the minimum required ductwork with wet feed chute to set the burner adjacent to the drum dryer inlet (optional extra ductwork available). Proposed fuel to be virgin wood sawdust, or approved by MCE, with a moisture content less than 14% and maximum particle size of 1/4".

Burner control panel accepts 460v 3φ 60 Hz electrical supply (575/3/60 or 380- 415/3/50 readily available upon request) and provides motor controls and starters for excess air fan(s), fuel feed fan, fuel feed auger, and maypole-type bin agitator; with automatic modulation of firing rate based on process outlet temperature (or other appropriate process parameter). Per NFPA 85, separate safety controls monitor temperature of burner, process inlet, and process outlet for shutdown in case of temperature extremes. Spark detection & extinguishing system such as

Name	Quantity	Unit price	Unit discount	Net price
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Firefly, Flamex, GreCon, or HansenTek is highly recommended, and is required by safety codes.

Burner control panel has provision for interconnection to spark detection system and other control systems. Panel is manufactured in a facility which implements UL product certification, and is UL 508a labeled. Emissions compliance is the customer's responsibility, including contacting emissions standards authorities to determine requirements. Refractory-lined components must be protected from freezing, and non-freezing weather is required until heat-curing is completed during commissioning.

Design combustion efficiency.....	99.935%
Rated gross energy output.....	6.5 million BTU/hr (1,9 MW)
Burner Body diameter, at flanges.....	6' 4" (2,03 m)
Burner Body diameter, at sidewall.....	6' 2" (1,93 m)
Burner Body height overall.....	14' 6" (4,42 m)
Burner Body total weight (mass), approximate.....	15,800 lbs (7.167 kg)
Ductwork diameter, at flanges.....	3' 0" (0,91 m)
Ductwork diameter, at sidewall.....	2' 8" (0,81 m)
Typical maximum consumption of dry wood dust.....	903 lbs/hr (409 kg/hr)
Maypole Bin Agitator.....	5 HP (3,7 kW)
Fuel Feed Fan.....	3 HP (2,2 kW)
Excess Air Fans (3 each).....	total 2.25 HP (1,7 kW)
Control Instruments, (max.).....	2 kVA (2 kVA)

Options available at additional cost (priced upon request):

- Special refractory for high alkali ash fuels and/or severe duty
- Thicker lining or double lining of burner bottom section
- Double insulation of burner bottom section
- Stainless steel for one or more burner sections and/or duct sections
- Triplex fuel feed ports for fine texture fuels
- PLC-based touch screen HMI and remote monitoring capabilities
- Certification of control panel to comply with Canadian cUL / CSA, or European CE

Company representative, to supervise/assist installation and familiarize employees with operation and maintenance on a per diem basis, plus travel. Round-trip travel expenses are payable in advance, on-site work is payable at the end of each week. Customer is responsible for site preparations such as grading, concrete work, building construction &

modifications; for providing machinery (such as fork lifts and cranes) with operators to unload and place the equipment; and for electrical wiring and starters for motors (except those included with equipment). Customer is also responsible for providing qualified personnel, tools, and equipment to carry out the work under our direction. Work not completed within 90 days of equipment shipping will be billed at rates in effect at time of completion.

On-site work, all expenses included, per day M-F.....	\$ 2,750.00
On-site work, all expenses included, per day weekends/holidays.....	\$ 3,640.00
Travel portal-to-portal, plus expenses, per day.....	\$ 2,750.00

Billing summary: Starts on effective date

Spark Detection and Extinguish	1	\$19,676.00	—	\$19,676.00
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Spark Detection & Extinguishment

We are pleased to quote the Factory Mutual (FM) approved expandable Hansentek AN6400 control panel and

Name	Quantity	Unit price	Unit discount	Net price
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equipment to protect one zone. This system is designed to operate at 50 PSIG water pressure at the nozzle, which will generate a flow of 19 GPM per nozzle.

This means the water system must maintain a minimum of 50 PSIG at the nozzle while delivering 19 GPM to each nozzle.

The following equipment is included for Detect/Extinguish/Shut Down of one 14" dust pipe with 500° maximum air stream temperature:

1 Hansentek AN6400-4 spark detection system panel (NEMA 12) complete with keypad, display, power supply, charger and back up batteries and supervision of AC power, battery condition and ground faults. Monitors zone activity for reporting and archival storage of up to 2000 alarms with date and time of occurrence.

1 Microprocessor based Intelligent Zone Controller model 422 with built-in sensitivity checking with capacity for 4 spark detectors, one water flow and one heat detector inputs. Two supervised outputs for water nozzles and auxiliary equipment and two heavy-duty relays for abort dampers and system fan shutdown.

2 Outdoor, NEMA 4 addressable high temperature infra-red direct optics spark detectors with built-in Sensi-Check calibrated through the lens sensitivity checking. Includes two stainless steel-clad fiber optic bundles per detector to "look" into the duct. Cable and detector mounting hardware included.

1 Nozzle/Valve assemblies with solenoid-controlled water spray system and water strainer. Nozzle mounting hardware included.

1 Monitored water flow switch to generate a trouble signal if there is a spark and no water flow or water flow and no spark.

1 Alarm horn with strobe light for remote Annunciation.

Billing summary: Starts on effective date

Onsite Commissioning Assistance	1	\$13,800.00	—	\$13,800.00
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Onsite Commissioning Assistance

Includes FOUR days onsite with TWO days of travel

All travel expenses are included

Per Day ONSITE Rate (ONE PERSON) (ENGINEER) \$2,075

Applies to ONSITE commissioning work performed

Onsite Charge includes per diem, lodging, and labor

At least ONE full day charge is required

Cost is based on jobsite being located within the Continental US.

Per Day TRAVEL Rate (ONE PERSON) (ENGINEER) \$2,750

Applies to TRAVEL days required to arrive onsite from the Automation facility

Travel Charge includes mileage, per diem, lodging, travel labor, fuel,

shipment cost for work related material, and rental car fee

At least TWO travel days are required per project. Only exception is if the electrical installation and all travel, to and from, can be completed in one day.

Then ONE travel day will be assessed to the customer

Cost is based on jobsite being located within the Continental US.

Onsite Startup Assistance

Per Day Commissioning

- Onsite Assistance is designed to verify proper I/O signals, motor operation, and process automation.

- Scope of assistance will consist of verifying the full process control with all materials available.

- Electrical and Mechanical scope must be complete and in working order for commissioning to finish.

Name	Quantity	Unit price	Unit discount	Net price
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- The end customer must maintain a clear and unrestricted access to the job area and equipment at all times during our onsite.

- Area must be clean and free of debris

- Company accepts NO liability for the equipment being commissioned. Any warranty concerns will be directed to the equipment provider.

- Any lifts, fork trucks, or other equipment necessary will be the responsibility of the customer unless specifically quoted. Craft Automation will make an effort to request any needed equipment prior to the onsite. However, equipment may be needed same day in some instances.

- Customer must furnish, prior to arrival, all necessary information, schematics, process information, and related documents or incur potential delays.

- Any and all customer requested relocating, cleaning, rework, modifications, and or repairs not specifically tied to the original project scope will be subject to additional charges.

- Service rates apply to a door to door basis.

- Onsite assistance scheduled with less than (15) business days notice will be billed under a 1.3 multiplier. This includes travel, lodging and meals

At least three weeks notice of requested onsite date is required.

Additional charges may be assessed if less than three weeks notice is given in order to compensate for increased travel costs.

*** Internet Connection to panel may be required in order for ***

***any service or troubleshooting to be performed ***

Billing summary: Starts on effective date

Subtotal	\$786,761.00
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Discount included in the subtotal	\$0.00
=====	
Due on effective date	\$786,761.00
The total amount due on the effective date, not including any future payments due later.	
=====	
Total contract value	\$786,761.00

Terms

Payments

Customer must agree to the following payment terms: due on receipt

Acceptance

This section is designated for the authorized representatives from both parties involved in this agreement. By signing below, the signatories confirm their acceptance of the terms and conditions outlined in this document.

On behalf of the buyer,

[sig|req|signer1]

Jim Nix

jimnix911@yahoo.com